

Collocation Analysis Harrisburg and Philly

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1. Introduction

In this analysis we calculate the r-squared values and root mean square errors of the low-cost sensors used in collocation in Harrisburg and Philadelphia. The goal is to understand how the data collected using the three air sensor types (airgradient, quantaq and clarity) compares to the one collected using EPA monitors and reported to airnow. By tabulating the results we can easily make analyses to understand how the devices perform when compared to EPA monitors.

Let's first set up some libraries that are likely to be useful

2. Data extraction from sensors

The data is extracted from sensors, some cleaning done on it before it is sent to a cloud hosting database known as influxdb. In this notebook we shall read data from influxdb and analyze it from here.

The sensor data from all the three collocation sites was downloaded, cleaned in python and upload to the shared drive. Below we read the csv that stores the data from these sensors.

We make sure that the data types for each attribute in the dataset is set to the expectation

```
'data.frame': 56565 obs. of 9 variables:
 $ X          : int  0 1 2 3 4 ...
 $ time_US_eastern : chr  "2026-04-15 20:00:00" "2026-04-15 21:00:00" ...
 $ latitude     : num  40.2 40.2 ...
 $ longitude    : num  -76.8 -76.8 ...
 $ device_sn    : chr  "MOD-X-00958" "MOD-X-00958" ...
 $ manufacturer  : chr  "quantaq" "quantaq" ...
 $ site_name    : chr  "Harrisburg" "Harrisburg" ...
 $ parameter    : chr  "PM2.5" "PM2.5" ...
 $ pm_gases_measures: num  7.29 7.27 ...
```

R doesnot seem to recognize the time column as time; so we convert it to time datatype for easy use during analysis

	time_US_eastern	latitude	longitude	device_sn	manufacturer
46298	2026-04-14 20:00:00	40.24701	-76.84699	840420430401	EPA
46322	2026-04-14 20:00:00	40.24701	-76.84699	840420430401	EPA
46299	2026-04-14 21:00:00	40.24701	-76.84699	840420430401	EPA

	site_name	parameter	pm_gases_measures
46298	Harrisburg	OZONE	36.0
46322	Harrisburg	PM2.5	4.8
46299	Harrisburg	OZONE	33.0

3. Data restructuring for RMSE and R-Squared

Let's filter out for only the devices we plan to use in this analysis. For consistency, we can use the devices that justair is using to calculate the collocation factors.

In addition, we shall also add the EPA monitors since they will be used for reference. Lets first identify the sn for the EPA monitors

Since we need to use 24 hour average data for pm2.5, we can average the data on 24hour

4. Calculating the r-squared and rmse

We can first make a table with all the thresholds recommended by epa for comparison

For each location we find the r-squared and rmse using the EPA data as reference for each low-cost 'representative' device per manufacturer. For reusability and simplicity,let's write a function that directly makes dataframes for r-squared and rmse based on paramter and location

We section the calculations by location for organization.

4.1 Harrisburg

The EPA monitors in Harrisburg only measure PM2.5 and Ozone.

4.1.1 Harrisburg PM2.5

We use the function to find the pm2.5 error values in Harrisburg for the three low cost sensors

	Location	Device.sn	Manufacturer	Parameter	R.squared
1	Harrisburg	MOD-X-00958	quantaq	PM2.5	0.97
2	Harrisburg	A93NM49L	clarity	PM2.5	0.957
3	Harrisburg	airgradient:588c8135a6ec	airgradient	PM2.5	0.942
	P.values	Target.R.squared	rmse	Target.rmse	
1	2.139003e-47	>=0.70	0.5	<=7µg/m ³	
2	1.195538e-42	>=0.70	0.6	<=7µg/m ³	
3	1.05461e-25	>=0.70	0.8	<=7µg/m ³	

Based on the values above, we can say that the R.squared and rmse values for all the devices (quantaq, clarity and airgradient) are within the EPA recommended range

4.1.2 Harrisburg Ozone

	Location	Device.sn	Manufacturer	Parameter	R.squared
1	Harrisburg	MOD-X-00958	quantaq	OZONE	0.723
2	Harrisburg	A93NM49L	clarity	OZONE	0.752
3	Harrisburg	airgradient:588c8135a6ec	airgradient	OZONE	0.632
	P.values	Target.R.squared	rmse	Target.rmse	
1	0e+00	>=0.80	6.3	<=5ppb	
2	0e+00	>=0.80	6	<=5ppb	
3	4.744426e-214	>=0.80	7.3	<=5ppb	

The r-squared for Ozone in Harrisburg are low below the EPA recommended threshold. Agradient seems to be having the lowest r-squared value and largest rmse

4.2 MON Philadelphia

The Philadelphia Montemogrey site measure PM2.5, CO and NO2.

4.2.1 MON - PM2.5

	Location	Device.sn	Manufacturer	Parameter	R.squared
1	Philly-MON	A636FW6Q	clarity	PM2.5	0.942
2	Philly-MON	airgradient:588c813fe180	airgradient	PM2.5	0.92
	P.values	Target.R.squared	rmse	Target.rmse	
1	3.525698e-23	>=0.70	1.3	<=7µg/m ³	
2	6.928558e-17	>=0.70	1.5	<=7µg/m ³	

Both the air gradient and Clarity at the Montgomery have an r-squared value that is greater than the epa recommended threshold . The rmse is also less than the threshold

4.2.1 MON - NO2

	Location	Device.sn	Manufacturer	Parameter	R.squared
1	Philly-MON	A636FW6Q	clarity	NO2	0.23
2	Philly-MON	airgradient:588c813fe180	airgradient	NO2	0.356
	P.values	Target.R.squared	rmse	Target.rmse	
1	5.345159e-45	>=0.70	4.5	<=5ppb	
2	1.536529e-61	>=0.70	4.1	<=5ppb	

The r-squared values for NO2 at the Montgomery site are way below the thrshold!

4.2.1 MON - CO

	Location	Device.sn	Manufacturer	Parameter	R.squared	P.values
1	Philly-MON	A636FW6Q	clarity	CO	0.798	2.429596e-275
	Target.R.squared	rmse	Target.rmse			
1	>=0.80	0	<=1ppm			

The CO value for the clarity at Montgomery site is just right on the threshold!

4.3 NEW - Philadelphia

4.3.1 NEW - PM2.5

	Location	Device.sn	Manufacturer	Parameter	R.squared
1	Philly-NEW	A44MFTF3	clarity	PM2.5	0.663
2	Philly-NEW	MOD-X-00959	quantaq	PM2.5	0.741
3	Philly-NEW airgradient:588c81359ef8	airgradient		PM2.5	0.637
	P.values	Target.R.squared	rmse	Target.rmse	
1	4.785889e-09	>=0.70	1.6	<=7µg/m ³	
2	1.331475e-10	>=0.70	1.4	<=7µg/m ³	
3	3.595473e-07	>=0.70	1.7	<=7µg/m ³	

Surprisingly, the pm2.5 r-squared values on the NEW site are below the threshold. This makes this site the only one where the pm2.5 r-squared values do not meet the threshold. The values are even lower when hourly data is used

4.3.2 NEW - OZONE

	Location	Device.sn	Manufacturer	Parameter	R.squared
1	Philly-NEW	A44MFTF3	clarity	OZONE	0.824
2	Philly-NEW	MOD-X-00959	quantaq	OZONE	0.89
3	Philly-NEW airgradient:588c81359ef8	airgradient		OZONE	0.728
	P.values	Target.R.squared	rmse	Target.rmse	
1	3.673595e-300	>=0.80	5.4	<=5ppb	
2	0e+00	>=0.80	4.3	<=5ppb	
3	1.025456e-184	>=0.80	6.9	<=5ppb	

Only airgradient slightly fails to reach the threshold for the r.squared value for ozone at the NEW site

4.3.3 NEW - NO2

	Location	Device.sn	Manufacturer	Parameter	R.squared
1	Philly-NEW	A44MFTF3	clarity	NO2	0.277
2	Philly-NEW	MOD-X-00959	quantaq	NO2	0.203
3	Philly-NEW airgradient:588c81359ef8	airgradient		NO2	0.588
	P.values	Target.R.squared	rmse	Target.rmse	
1	2.019119e-57	>=0.70	4.8	<=5ppb	
2	8.03833e-39	>=0.70	5.1	<=5ppb	
3	2.949638e-127	>=0.70	3.8	<=5ppb	

The r.squared for NO2 at the MON site are all below the threshold. Surprisingly only air gradient is within close to 0.7 and it has the lowest rmse

4.3.4 NEW - CO

	Location	Device.sn	Manufacturer	Parameter	R.squared	P.values
1	Philly-NEW	A44MFTF3	clarity	CO	0.481	1.154292e-111
2	Philly-NEW	MOD-X-00959	quantaq	CO	0.311	2.646471e-61
	Target.R.squared rmse Target.rmse					
1		>=0.80	0	<=1ppm		
2		>=0.80	0	<=1ppm		

Both clarity and quantaq CO r-squared values at new are below the threshold with clarity having a slightly higher r-squared

5. Summary table of all parameters at all locations

	Location	Device.sn	Manufacturer	Parameter	R.squared
1	Harrisburg	MOD-X-00958	quantaq	PM2.5	0.97
2	Harrisburg	A93NM49L	clarity	PM2.5	0.957
3	Harrisburg	airgradient:588c8135a6ec	airgradient	PM2.5	0.942
4	Harrisburg	MOD-X-00958	quantaq	OZONE	0.723
5	Harrisburg	A93NM49L	clarity	OZONE	0.752
6	Harrisburg	airgradient:588c8135a6ec	airgradient	OZONE	0.632
7	Philly-MON	A636FW6Q	clarity	PM2.5	0.942
8	Philly-MON	airgradient:588c813fe180	airgradient	PM2.5	0.92
9	Philly-MON	A636FW6Q	clarity	NO2	0.23
10	Philly-MON	airgradient:588c813fe180	airgradient	NO2	0.356
11	Philly-MON	A636FW6Q	clarity	CO	0.798
12	Philly-NEW	A44MFTF3	clarity	PM2.5	0.663
13	Philly-NEW	MOD-X-00959	quantaq	PM2.5	0.741
14	Philly-NEW	airgradient:588c81359ef8	airgradient	PM2.5	0.637
15	Philly-NEW	A44MFTF3	clarity	OZONE	0.824
16	Philly-NEW	MOD-X-00959	quantaq	OZONE	0.89
17	Philly-NEW	airgradient:588c81359ef8	airgradient	OZONE	0.728
18	Philly-NEW	A44MFTF3	clarity	NO2	0.277
19	Philly-NEW	MOD-X-00959	quantaq	NO2	0.203
20	Philly-NEW	airgradient:588c81359ef8	airgradient	NO2	0.588
21	Philly-NEW	A44MFTF3	clarity	CO	0.481
22	Philly-NEW	MOD-X-00959	quantaq	CO	0.311
	P.values Target.R.squared rmse Target.rmse				

1	2.139003e-47	>=0.70	0.5	<=7µg/m ³
2	1.195538e-42	>=0.70	0.6	<=7µg/m ³
3	1.05461e-25	>=0.70	0.8	<=7µg/m ³
4	0e+00	>=0.80	6.3	<=5ppb
5	0e+00	>=0.80	6	<=5ppb
6	4.744426e-214	>=0.80	7.3	<=5ppb
7	3.525698e-23	>=0.70	1.3	<=7µg/m ³
8	6.928558e-17	>=0.70	1.5	<=7µg/m ³
9	5.345159e-45	>=0.70	4.5	<=5ppb
10	1.536529e-61	>=0.70	4.1	<=5ppb
11	2.429596e-275	>=0.80	0	<=1ppm
12	4.785889e-09	>=0.70	1.6	<=7µg/m ³
13	1.331475e-10	>=0.70	1.4	<=7µg/m ³
14	3.595473e-07	>=0.70	1.7	<=7µg/m ³
15	3.673595e-300	>=0.80	5.4	<=5ppb
16	0e+00	>=0.80	4.3	<=5ppb
17	1.025456e-184	>=0.80	6.9	<=5ppb
18	2.019119e-57	>=0.70	4.8	<=5ppb
19	8.03833e-39	>=0.70	5.1	<=5ppb
20	2.949638e-127	>=0.70	3.8	<=5ppb
21	1.154292e-111	>=0.80	0	<=1ppm
22	2.646471e-61	>=0.80	0	<=1ppm